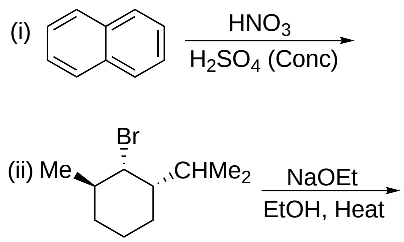
ACML 101: Organic Chemistry Assignment 2 Solutions

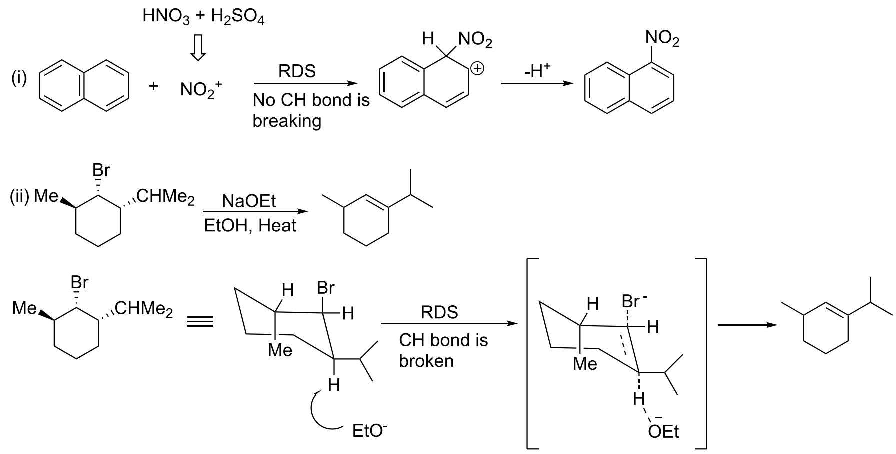
Q1. Arrange the following in increasing order of their relative acidity.

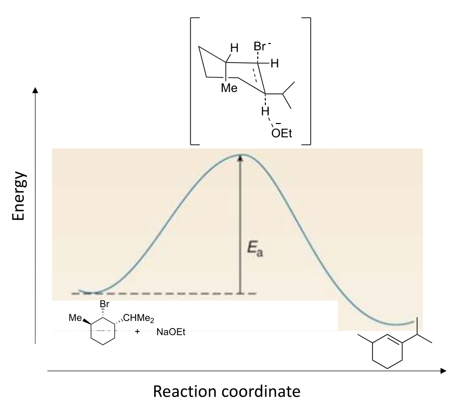
Answers at the last page.

Q2. Which of the following reactions will show kinetic hydrogen isotope effect? Explain with the help of mechanism and identify the RDS. Give the product formed in reaction (ii) and draw the labeled energy profile diagram.

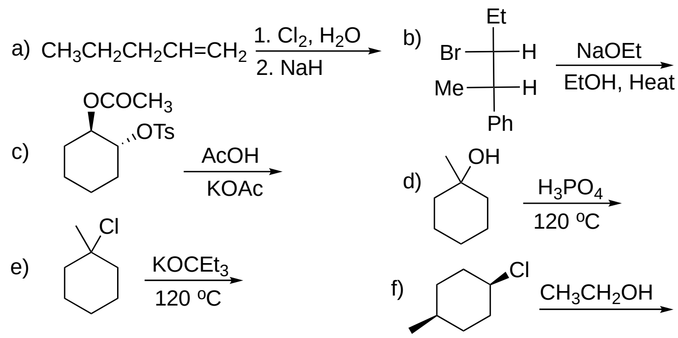


Ans 2. CH bond is breaking in the RDS step in (ii) hence, this reaction will show the kinetic hydrogen isotopic effect.

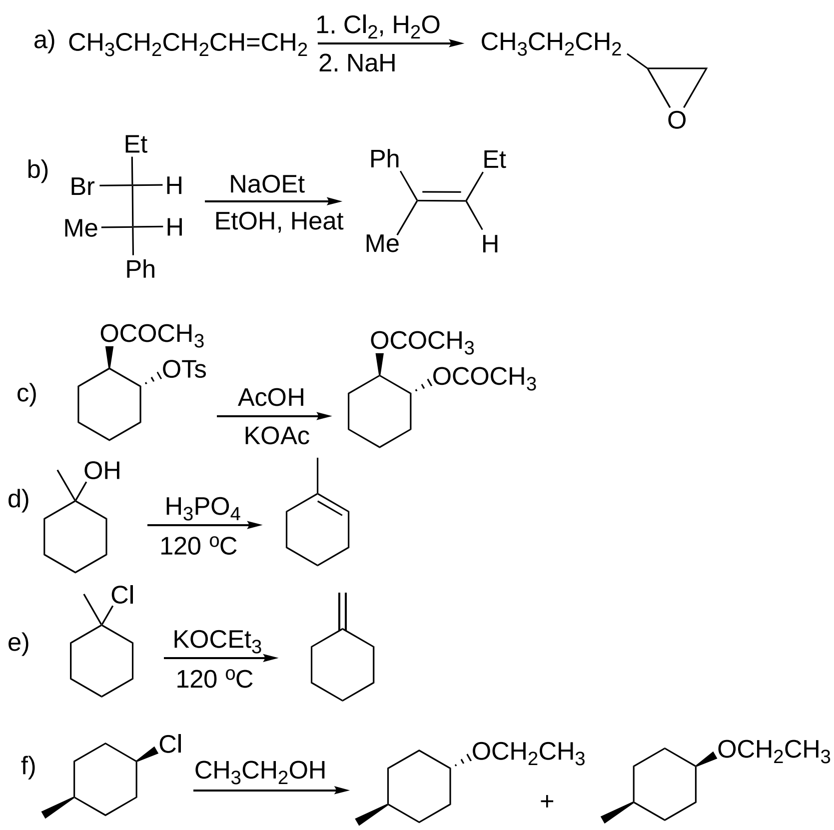




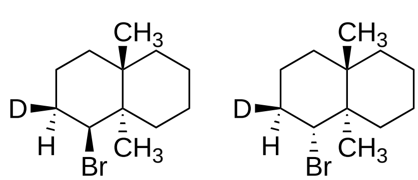
Q3. Identify the major product in the following reactions with correct stereochemistry wherever applicable.



Ans 3.

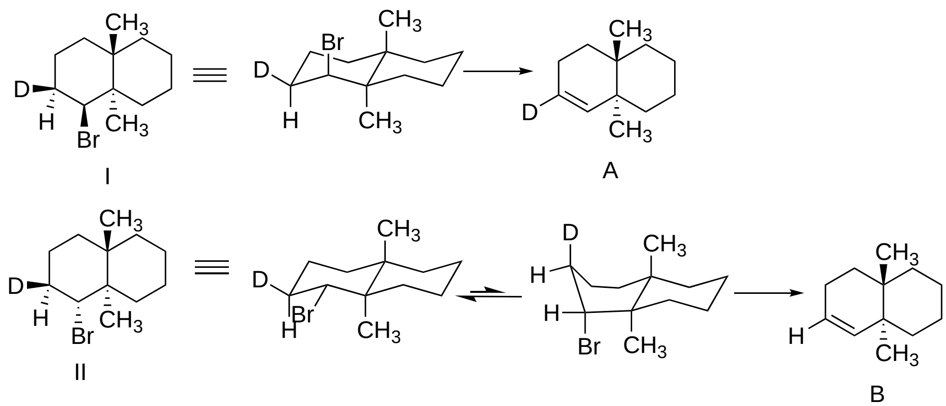


Q4. On attempted E2 reaction, (I) reacted much faster than (II). Write the product of the reaction in each case and explain the reason.



I II

Ans 4. The stereoelectronic requirement of an E2 reaction is that the eliminating groups should be antiperiplanar. In cyclohexane ring this relationship exists when substituents on the two adjacent carbons are in axial positions. Hence, product A will be oobtained from I and B will be obtained from II. The difference in the observed rates is due to the cleavage of C-H vs C-D bond in the RDS in either reaction.



Q5. For the S_N2 rection,

$$CH_3CH_2CH_2CH_2Br + NaOH \xrightarrow{\text{Ethanol}} CH_3CH_2CH_2CH_2OH + NaBr$$

Answer the following questions:

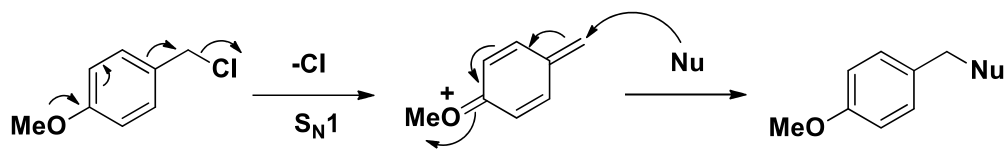
(a) What is the rate expression for the reaction?
(b) Draw an energy diagram for the reaction. Label all parts. You may assume that the products are of lower energy than the starting reagents.
(c) What will be the effect on the rate of reaction if the concentration of *n*-butyl bromide is doubled?
(d) What will be the effect on the rate of rection if the concentration of the NaOH is halved?
(e) How will the rate of the SN_2 reaction change if the solvent is changed to 80% ethanol and 20% water?

Answers at the last page.

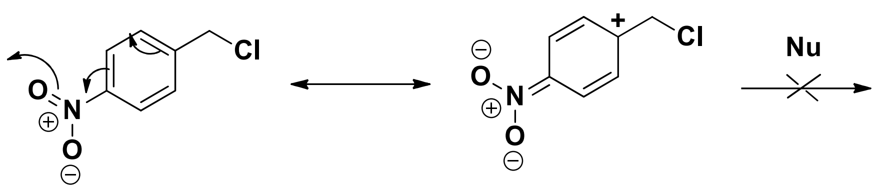
Q6. Predict the reaction mechanism and compare the rates in two different sets of reactions.

Ans 6.

A benzylic chloride that favours the S_N1 mechanism

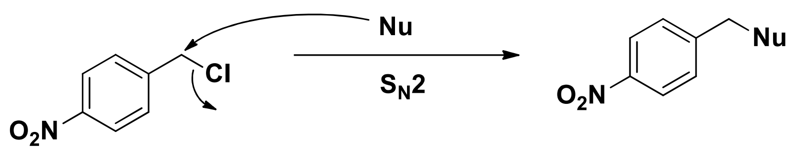


But a benzylic chloride that disfavours the S_N1 mechanism

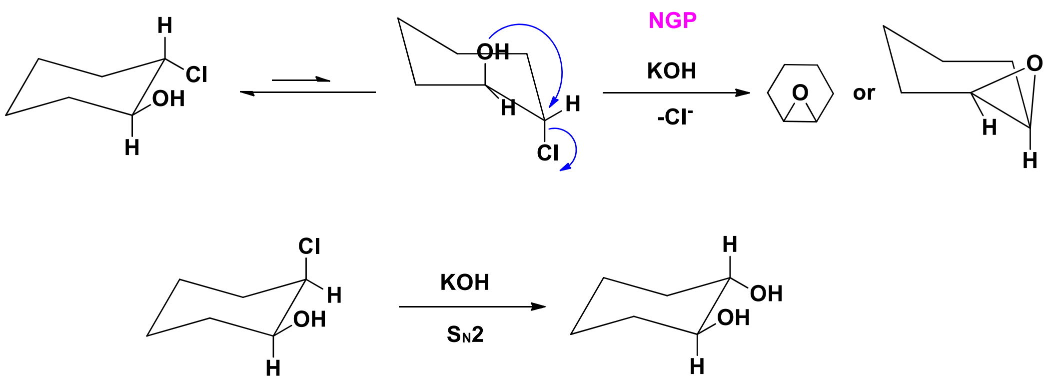


If we set the rate of substitution of the benzyl compound with methanol at 25 °C, then the 4-MeO benzyl compound reacts about 2500 times faster and the 4-NO_2 benzyl compound.

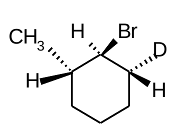
The same benzylic chloride that favours the S_N2 mechanism



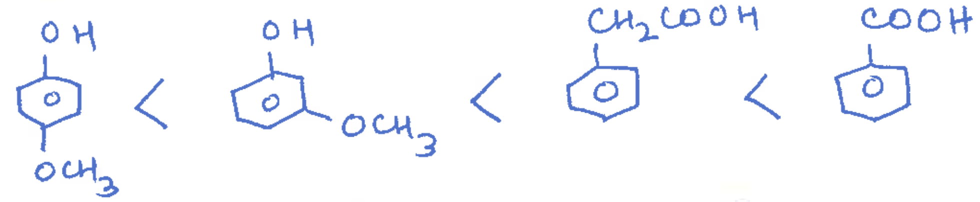
Q7. Predict the product(s) structure in the following two sets of reactions.
Ans 7.

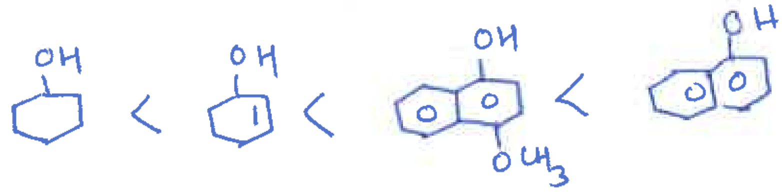


Q8. When the deuterium labelled compound shown below is subjected to dehydrohalogenation using alcoholic base, the only alkene product is 3-methylcyclohexene with no deuterium.
Answer at the last page.



Answer 1 :

a) 

b) 

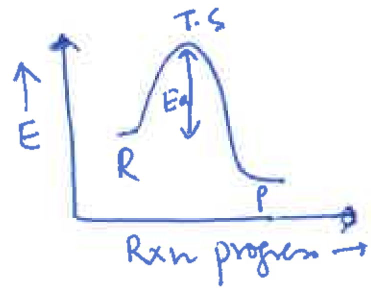
c) $CH_3OH < HCN < HOAc < HOTs$

d) $HOOC-CH_2-COO^- < CH_3COOH < HOOC-CH_2-COOH < O_2N-CH_2COOH$

e) $CH_3CH_2COOH < H_3CO\,CH_2COOH < Me_3\overset{+}{N}CH_2COOH < O_2N-CH_2COOH$

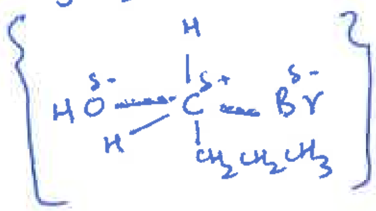
Ans 5 :

a) $Rate = k[CH_3CH_2CH_2CH_2Br][OH^-]$

b) 

$R = [CH_3CH_2CH_2CH_2Br] + [NaOH]$

$P = [CH_3CH_2CH_2CH_2OH] + [NaBr]$

T.S. = 

c) Rate doubles

d) Rate is halved

e) Reaction becomes slower as Nucleophile gets solvated and its reactivity decreases.

Ans 8.

Product :